

UNIVERSITY OF MAURITIUS

FACULTY OF LAW AND MANAGEMENT



FIRST SEMESTER EXAMINATIONS

FEBRUARY 2021

PROGRAMME	BSc (Hons) Management (Minor: Business Informatics) Level III		
MODULE NAME	INFORMATION SYSTEMS SECURITY		
DATE	Monday 15 February 2021	MODULE CODE	CSE3216(3) / ICT3095
TIME	09:30 – 11:30 Hours	DURATION	2 Hours
NO. OF QUESTIONS SET	5	NO. OF QUESTIONS TO BE ATTEMPTED	4

INSTRUCTIONS TO CANDIDATES

There are TWO (2) Sections in this paper : Section A and Section B.

Section A is COMPULSORY and carries 40 marks.

**Section B consists of FOUR (4) questions. Answer ANY THREE (3) questions.
Each question carries 20 marks.**

SECTION A : COMPULSORY [40 marks]

Question 1 [40 Marks]

- a) Explain the difference between *security vulnerability* and a *security threat* using appropriate examples.
[3 x 2 marks]
- b) Differentiate between active and passive attack.
[3 x 2 marks]
- c) Briefly explain the following terms and measures to counteract each of these attacks:
(i) Denial of Service
(ii) Ransomware
(iii) Phishing
(iv) Zero-day attacks
[4 x 4 marks]
- d) Discuss the advantages and disadvantages for organisations to host their applications on the cloud environment.
[6 x 2 marks]

SECTION B : ANSWER ANY THREE (3) QUESTIONS

Question 2 [20 Marks]

- a) Explain by using an appropriate example of the following cryptographic terms:
(i) hashing
(ii) digital certificate
[3 + 3 marks]
- b) Compare and contrast symmetric key encryption to asymmetric key encryption in terms of:
(i) Complexity and speed of execution
(ii) Security
[2 x 3 marks]
- c) Discuss how cryptography can address organisation security requirements in providing security.
[4 x 2 marks]

Question 3 [20 Marks]

- a) Explain, by providing three reasons, why security is an important aspect in wireless networks?
[3 x 2 marks]
- b) Describe three common types of wireless attacks and measures to counteract each one of these attacks.
[6 marks]
- c) Discuss four different security requirements that organisation should enforce in wireless networks.
[8 marks]

Question 4 [20 Marks]

- a) What is an SQL injection attack and describe the different forms of mechanisms that can be deployed to prevent such an attack.
[6 marks]
- b) What is cross-site scripting (XSS) and describe how it can be prevented.
[2+ 4 marks]
- c) Explain how the use of security policies can help organisations to be much safer.
[8 marks]

Question 5 [20 Marks]

- a) Differentiate between stateful and deep packet firewalls.
[6 marks]
- b) Explain the difference between signature and anomaly detection in intrusion detection systems (IDSes).
[6 marks]
- c) Describe the term Business Continuity Planning (BCP) and list the different key components of BCP.
[8 marks]

END OF QUESTION PAPER